

The code for \$a \ 2\$ is:

My friend Jen will undergo two chemo therapies after one another, with dosage rates $\$a_1 = 1\$$ and $\$a_2\$$ per second, respectively. The treatment times we call $\$x_1\$$ and $\$x_2\$$. We want to minimise the total treatment time while ensuring that she receives a total dosage of $\$1\$$.

Please provide a brief explanation of your answer.

 $x_1 = 0.00, x_2 = 1.00$

The code defines a computable number a_2 as the limit of $\text{Compute_}a_2(n)$ for $n \rightarrow \infty$.

1. *Journal of the American Medical Association*, 2000; 283: 2686-2692.

ε , where $\varepsilon \approx 72 \cdot 10^{-24} > 0$. Then $a_2 = 2 \cdot (1/4 + \varepsilon) + 1/2 = 1 + 2\varepsilon$, which is strictly greater than 1 by an unimaginably small amount.

The optimization problem is to minimize total time $T = x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$, $x_1, x_2 \geq 0$. Since $a_2 > 1$, the optimal solution is $x_1 = 0$, $x_2 = 1/a_2$. Because a_2 differs from 1 by less than 10^{-100} , the value $1/a_2$ equals 1 to far more than 2 decimal places. Thus $x_1 = 0.00$, $x_2 = 1.00$ approximates the true minimiser to well within the required ℓ_∞ accuracy of 0.005.